

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	HAZOP Terms of Reference
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HAZOP Terms of Reference

R01	05.07.22	Issued for Review	V. Orafu	E. Maduabuchi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	05-07-22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-011

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
DEP	Design and Engineering Practice
F & G	Fire & Gas
HAC	Hazardous Area Classification
HAZOP	Hazard and Operability Studies
PEFS	Process Engineering Flowscheme
PFS	Process Flowscheme

1. Introduction

1.1 Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

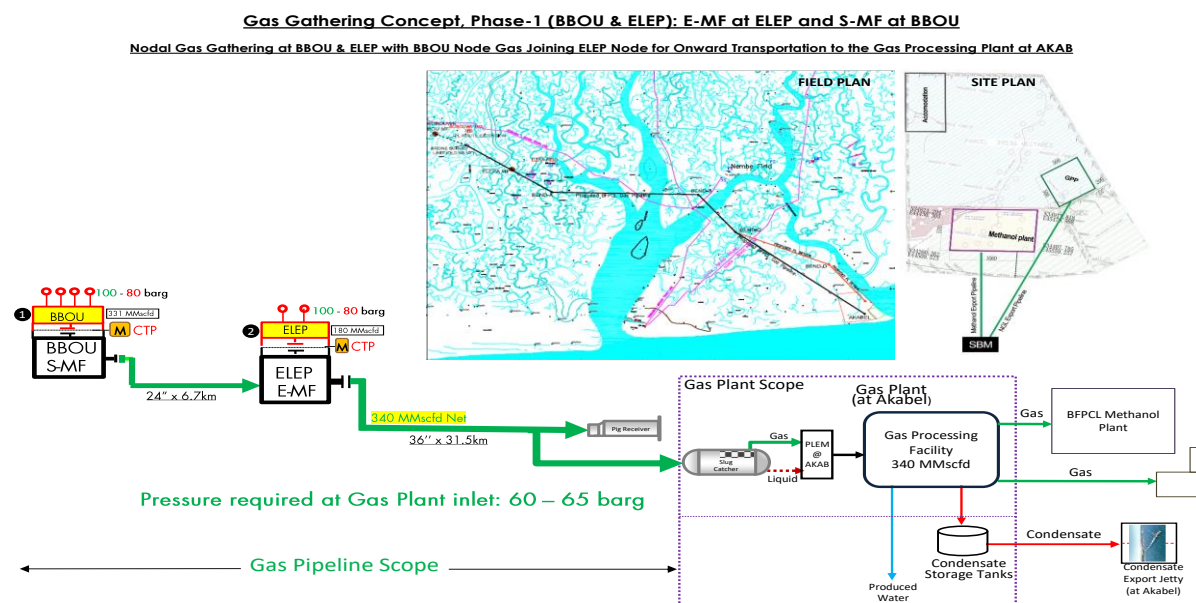


Figure 1: Gas Gathering Concept, Phase 1

1.2 Purpose

The purpose/objectives of the HAZOP study are to:

- Assess risks and control measures in the design, to ensure risks are adequately managed to ALARP
- Recommend suitable actions towards risk reduction if the controls and recovery measures are not deemed to be sufficient to control the hazard
- Identify operability and maintainability issues in the design
- Prepare a report of the findings of the study

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.00.00.15-Gen	Hazard and Operability (HAZOP) Study

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Methodology

A HAZOP is a structured analysis of a process, system or operation carried out by a multi-discipline team of competent people and chaired by an independent person. The team carries out a systematic examination of the processes and operations of the facilities covered within this Terms of Reference. This is aided by using a set of guidewords in combination with parameters to give meaningful deviations from the design intent. The team concentrates on those deviations that could lead to:

- Potential hazards to safety, health, or the environment; and
- Operability problems.

Where realistic causes of deviation are found, the team evaluates the consequences based on the team's experience and judgment. If the existing safeguards are deemed inadequate, the team will record an action for change or recommend further investigation of the problem.

3.1 Guidewords

HAZOP guidewords (Deviation + Parameter) shall be adopted for the review session. Typical guidewords are shown as Appendix 1. The guidewords are intended to prompt discussion and are not intended to be exhaustive

3.2 Preparation Work, Nodes, and Documentation

A preliminary note delineation has been carried out by the Facilitator (ESLP Consultants Limited). The proposed nodes of study are in Table 1 below. The marked-up drawings are shown in Appendix 2

Table 1: Proposed HAZOP Nodes

Node	Title	Drawing No.
1	BBOU MF Receiver & Launcher	EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_1 R01; EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_2 R01
2	BBOU Open Drain & Closed Systems and Vent System	EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_3 R01
3	ELEP MF Receivers & Launcher	EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_4 R01; EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_5 R01; EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_6 R01
4	ELEP Open Drain, Closed Drain System and Vent System	EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_7 R01
5	AKABEL Pig Receiver	EA-BFPCL-NGC-GPMC-GGPL-FEED-PSE-DWG-003_8 R01

3.3 Preparation Work, Nodes, and Documentation

The following key documentation will be required for the HAZOP study

- Approved for HAZOP Process Flow Diagram(s)
- Approved for HAZOP Process Engineering FlowScheme (PEFS)
- Approved for HAZOP Process Control Narratives & Safeguarding Memorandum
- Plot plan
- Flowing and Closed In Tubing Head Pressure of the NAG wells at BBOU and ELEPA

Other documents to be kept by the Design Team for reference purposes

- Process and F & G Cause & Effect Matrices
- Process Safeguarding narratives
- Materials selection report(s)
- Vent dispersion calculation report
- Fire & Explosion Risk Assessment reports
- Hazardous Area Classification (HAC) drawings
- Operations Manual

3.4 HAZOP Agenda

The agenda will be as follows:

- Team introduction, domestic arrangements, safety instructions
- HAZOP Chairman (Facilitator) introduce the scope of the HAZOP study
- Design Team Process Engineer presents the process description of the project, highlighting the key features and design decisions taken to date
- The HAZOP Chairman will explain the HAZOP process including the HAZOP worksheet format and terminologies
- The Chairman will:
 - Propose Nodes and agree on Nodes with the team.
 - Commence HAZOP using the guidewords. This is not a rigid list and the team can add additional hazards not covered by these guidewords as required.

- All relevant information will be recorded on the worksheet.
- Refreshments & comfort breaks shall be held at intervals as agreed by the team.

3.5 HAZOP Session Details

The HAZOP will be carried out on the 8th of July 2022, commencing at 09.00 AM and finishing by 5:00 PM same day, virtually. Attendees/ reps will be shown as in Table 2 below.

Table 2: HAZOP Attendee List

Discipline	Name	Company
ESLP Consultants Reps		
HAZOP Chairman	Engr. Emeka Maduabuchi	ESLP Consultants Ltd
Scribe	Engr. Orafu Valentine	ESLP Consultants Ltd

Other Design Team disciplines – Mechanical/Piping, Materials, Civil/Structural, Electrical, etc. are not core members of the HAZOP team and therefore will be on standby

The HAZOP shall be recorded using PHApro 8.5.1.0 Software. The worksheets shall be projected onto a screen so that team members have an opportunity to review and comment during the session. Actions recorded during the session shall be assigned to individual team members for follow-up and closeout after the meeting

3.6 Deliverables & Follow Up Work

A HAZOP worksheet presenting a record of the HAZOP findings will be issued to all attendees for review to confirm that the report reflects the session content. A HAZOP report will be issued to the HAZOP Co-ordinator, with the HAZOP worksheet as an attachment to the report

Action close out methodology will be as follows:

- Action sheets will be sent to the client's focal point who will, in turn, send all actions to identified actionees for effective action closeout
- The respondee will give the action response to the Design Team Discipline Lead/TA for sign-off before returning the form to the Design Team Technical Safety/HSE engineer for formal close out.

A HAZOP action close-out report will be issued by the Design Team once all actions have been addressed and closed out by the Technical Safety/HSE engineering team.

Appendix 1: Guidewords

GUIDEWORD	PROPERTY	POSSIBLE CAUSES
NO	Flow	Wrong routing, blockage (debris, hydrate, sand), slip blind left in, faulty non-return valve, burst pipe, control valve, isolation valve, pump or vessel failure.
MORE	Flow	More than 1 pump operating, reduced delivery head, increased suction pressure, lower downstream pressure, other routes, exchanger tube leaks, greater density, gas break through, valve malfunction.
	Pressure	Surge, relief, leakage from an HP connection (lines and flanges), thermal, rate of pressurising lines, blockage downstream.
	Level	Surge, blocked outlet, control valve failure (see also More Flow)
	Temperature	Fouled cooler, cooling water failure, failed exchanger tubes, external fire.
	Viscosity	Change of material, specification or temperature.
LESS	Flow	Partial blockage, vessel or valves failing, leaks, loss of pump efficiency.
	Pressure	Generation of vacuum by pump or drain out of vessels, condensation from vapour or gas dissolving in a liquid. Blocked pump or compressor suction.
	Level	Leaks, control valve failure, upstream upset, blockage
	Temperature	Freezing, loss of pressure (Joule Thompson effect), loss of heating, failed exchanger tubes.
	Viscosity	Change of material, specification, or pressure.
REVERSE	Flow	As for NO FLOW, plus emergency venting, 2-way flow.
PART OF	Composition change	Passing through isolations, leaking exchanger tubes, wrong feed material. Phase change, out of specification.
AS WELL AS	Contamination	H ₂ S, mercury, radioactive nuclides, wax, sand, oxygen in water.
OTHER THAN	Relief	Relief valve, bursting disc, location of discharge, bellows type, fire.
	Sampling	Type, cooling of, response time.
	Corrosion and Erosion	Material specifications, fluid velocities, baffles, pipe-guides, hangers, anchors, external attack, sulphide/chloride stress corrosion, zinc embrittlement.
	Service Failures	Cooling Water, instrument air, steam, nitrogen or inert gas; power to machines, control circuits and computers.
	Maintenance	Isolation, drainage, purging, cleaning, drying, access, incorrect installation of equipment, hydrotest, lifting.
	Static	Electrical grounding, low conducting liquids, insulated conductors.
	Outside conditions	Fire, weather, environmental change, impact flood, etc.
	Operation	Start up, shut down (normal, emergency), maloperation, pigging, layout, different modes.
	Layout	Ergonomics issues, (raised once per unit)

[illegible]

